Deep Learning Architectures for RSSI-Based Passenger Movement Classification in Public Transport

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***Abstract*—Accurate, non-intrusive monitoring of passenger flows is essential for intelligent public transport systems. Prior work demonstrated that temporal sequences of Wi-Fi Received Signal Strength Indicator (RSSI) measurements can classify passenger movements using classical machine learning, achieving a Matthews Correlation Coefficient (MCC) of 0.756 on combined datasets. This paper extends that line of research by systematically evaluating deep learning architectures—including recurrent, convolutional, transformer, state-space, and mixture-of-experts models—for the same classification task across four dataset variants. We introduce 11 novel architectures, a class-conditioned data augmentation strategy expanding 2,253 samples to 10,115, and comprehensive hyperparameter optimization with 100 Optuna trials per model. Our best ensemble model, a two-stage deep stacking architecture, achieves MCC of 0.859 and accuracy of 89.4% on combined data, a 13.6% relative improvement over classical baselines. We further analyze the impact of multi-device interference, per-class confusion patterns, and the effectiveness of metadata-aware architectures in distinguishing true movement from environmental noise.**

***Index Terms*—Passenger counting, RSSI fingerprinting, Wi-Fi sensing, public transport, deep learning, state space models, mixture of experts, hyperparameter optimization**

I. INTRODUCTION

Public transport systems serve over 75% of EU urban citizens, yet inefficiencies contribute to 24% of greenhouse gas emissions. Accurate passenger flow knowledge—boarding, alighting, and occupancy per stop—is fundamental to network optimization and dynamic scheduling.

Traditional Automatic Passenger Counting (APC) relies on cameras, infrared sensors, or manual counting, each facing tradeoffs in cost, privacy, and deployment complexity. Wi-Fi sensing offers a privacy-preserving alternative: by analyzing Received Signal Strength Indicator (RSSI) patterns from passengers' devices, operators can infer movement without active user participation or additional hardware beyond existing access points.

However, RSSI measurements are inherently noisy, affected by multi-path propagation, device heterogeneity, and interference from concurrent transmissions. In prior work [1], we introduced a controlled dataset simulating four passenger movements—inside vehicle (AA), boarding (BA), alighting (AB), and at stop (BB)—and achieved MCC = 0.756 on combined data using Gaussian Processes from classical machine learning. This paper investigates whether modern deep learning architectures can better capture RSSI temporal dynamics, particularly under noisy multi-device conditions.

Our main contributions are:

1) Systematic evaluation of 11+ DL architectures—including state-space models (Mamba-3), mixture-of-experts, and two-stage stacking ensembles—across four dataset variants with increasing interference complexity.
2) A class-conditioned data augmentation strategy expanding 2,253 to 10,115 samples through targeted Gaussian noise injection and intra-class Mixup.
3) Comprehensive hyperparameter optimization via 100 Optuna trials per model spanning 19 search spaces.
4) Detailed interference analysis quantifying cross-route noise impact through heatmaps, trajectory visualization, and per-timestep perturbation measurements.

Our DeepStackEnsemble achieves MCC = 0.859—a 13.6% relative improvement over classical baselines. The remainder of this paper covers related work (Section 2), dataset and methodology (Section 3), experimental results with noise analysis (Section 4), and conclusions (Section 5).

II. RELATED WORK

Passenger counting spans multiple sensing modalities. Video-based APC using YOLOv5 achieves high accuracy but degrades under occlusion and variable lighting [2]; edge AI cameras address scalability at the cost of deployment complexity [3]. ML pipelines on traditional APC data enable offline correction of sensor errors [4], [5]. All camera-based methods raise privacy concerns and require substantial infrastructure.

Wi-Fi sensing avoids these drawbacks. iABACUS [6] and Myrvoll et al. [7] demonstrated counting via probe requests and signature analysis. Fabre et al. [8] applied RF and gradient boosting for ridership estimation. Oransirikul et al. [9] distinguished passenger from non-passenger activity, while Simoncic et al. [10] monitored presence through passive probes. CSI-based methods [11], [12] offer finer signal characterization but depend on specialized hardware unavailable in commodity

APs. RSSI fingerprinting for localization has been extensively surveyed [13]–[15]; BLE beacons [16] are an alternative but require dedicated infrastructure. Cerqueira et al. [17] addressed the complementary problem of OD matrix inference from smart card data.

Recent DL advances inform our architectural choices. Mamba [18] introduced selective state-space models with $O(L)$ complexity, extended by Mamba-2 [19] and Mamba-3 MIMO [20]. Autoformer [21] replaced self-attention with auto-correlation for time series. Mixture of Experts [22] enables conditional computation through sparse activation. Our prior work [1] established classical ML baselines (MCC = 0.756) on this dataset across 34 algorithms. This paper extends that analysis with 11+ specialized DL architectures, comprehensive HPO via Optuna [23], and systematic interference analysis.

III. Materials and Methods

A. Dataset Description

The dataset was collected in a controlled environment simulating public transport, as detailed in [1]. Zone A (room) represents the vehicle interior with an AP near the door; Zone B (corridor) represents the bus stop. Four devices executed four movement patterns—AA (inside vehicle), BB (at stop), BA (boarding), AB (alighting)—producing 10 RSSI readings at 1 s intervals. Two scenarios were collected: pure (single-device, 160 samples) and noise (four concurrent devices, 1,200 samples). The combined dataset contains 2,253 samples with class distribution AA (407), AB (387), BA (399), BB (417).

For DL input, each 10-dimensional RSSI sample is enriched into four channels per time step: raw RSSI r_t, velocity $\Delta_t = r_t - r_{t-1}$, acceleration Δ_t^2, and window-deviation $d_t = r_t - \mu_w$ (μ_w = per-sample mean).

B. Proposed Architectures

We evaluate 11+ architectures for 4-class classification.

Standard recurrent baselines include RNN, GRU [24], LSTM [25], and BiLSTM, each with 2 layers and hidden size 64. TCN [26] uses dilated causal convolutions with residual connections to capture long-range dependencies through exponentially increasing dilation. A Transformer with 4-head self-attention and sinusoidal position encoding enables global pairwise interactions across the 10-step window.

Beyond standard architectures, we propose several specialized models. The MambaClassifier adapts Mamba-3 MIMO [20] for sequence classification, leveraging the selective state-space mechanism for $O(L)$-complexity sequence modeling with dual-path CUDA-optimized kernels and rank-$R = 16$ input-dependent projections. CNN2DRNN treats (T, F) sequences as 2D images through Conv2d blocks with kernels $(3, 3)$ and $(5, 3)$, followed by LSTM/GRU temporal aggregation and attention pooling. MetaFusion fuses an LSTM/GRU backbone with learned embeddings (dimension 8) encoding the noise flag and concurrent interference path, enabling context-aware classification that distinguishes true movement from interference artifacts.

For multi-scale and ensemble approaches, a Multi-Scale CNN uses three parallel branches with kernel sizes $(3, 5, 7)$, each with residual blocks, concatenated through attention pooling. SoftMixtureOfExperts combines heterogeneous experts (GRU, LSTM, RNN, CNN, TCN) with learned top-2 gating and an auxiliary load-balance loss; 12 expert combinations were evaluated. Autoformer [21] replaces self-attention with FFT-based auto-correlation and adds series decomposition blocks, adapted for classification with a classifier head.

The DeepStackEnsemble is a two-stage meta-learner: 18 frozen base models (9 architectures $\times$ 2 configurations A/B) produce logit vectors, which feed per-class Level2Nets followed by a 3-layer residual DeepMetaMLP with skip connection. Joint fine-tuning at reduced learning rate further improves calibration.

C. Augmentation and Implementation

Two strategies expand training data from 2,253 to 10,115 samples. Class-conditioned Gaussian noise adds 6 copies to clean samples ($\sigma = 3.0$ dBm) and 4 copies to noisy samples ($\sigma = 0.8$ dBm), preserving each subset's noise characteristics. Intra-class Mixup blends 600 random pairs per class as $X_{\text{new}} = \lambda X_1 + (1 - \lambda) X_2$ with $\lambda \sim \text{Beta}(0.4, 0.4)$, restricted to same-label pairs.

The pipeline uses PyTorch 2.6 with Optuna 4.7. Training: AdamW, cosine annealing, early stopping (patience 25), label smoothing ($\varepsilon = 0.1$), training Mixup ($\alpha = 0.2$), L1 ($\lambda = 10^{-5}$), L2 (10^{-4}), dropout ($p = 0.3$), gradient clipping (1.0), batch 64.

HPO: 100 Optuna TPE trials/model searching architecture, training, loss, optimizer, and scheduler hyperparameters. SoftMoE uses NSGA-II. Primary metric: MCC [27], [28]. 5-fold stratified CV, 80/20 splits, 3 seeds. $\sim$65 models evaluated.

IV. Experimental Results

A. Experimental Setup

Four variants stress different robustness aspects: Pure (128/32 train/test, single-device), Noise (960/240, four concurrent devices), Normal (1,802/451, all original samples), Augmented ($\sim$9,875/$\sim$240, synthetic training). All models use 5-fold stratified CV, 80/20 splits; primary metric MCC [27], [28]. Seed 42 unless noted.

B. Variant-Level Comparison

Figure 1 shows best MCC per variant. Pure yields RNN 0.881 (seed 3). Noise degrades to DS RNN_B 0.798 (9.4% drop). Normal reaches DeepStackEnsemble 0.859; augmentation lifts HPO TCN to 0.865. Vs. classical ML [1] (GP 0.756 combined, KNN 0.922 pure, CatBoost 0.770 noise): DL improves combined (+13.6%) and noise (+3.6%), but trails on pure (−4.4%) where 128 samples are insufficient.

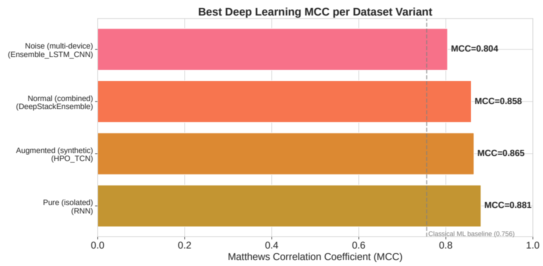

Figure 1. Best model MCC per variant. Classical baseline (0.756) dashed.

C. Architecture Performance and HPO

CNN (MCC 0.838) and TCN (0.836) lead single architectures; recurrent defaults underperform (RNN 0.755, GRU 0.710). Mamba (0.800) beats Transformer (0.713) by 12.2% with $O(L)$ complexity. HPO yields large recurrent gains: GRU 0.710→0.852 (+20.0%), LSTM 0.686→0.844 (+23.1%), BiLSTM 0.684→0.820 (+19.9%). CNN/TCN gain less. Each model underwent 100 Optuna TPE trials; SoftMoE used NSGA-II.

Table I lists top models. DeepStackEnsemble (18 base models → per-class Level2Net → residual MetaMLP) achieves best combined MCC 0.859. Voting pairs improve over individuals (CNN+TCN 0.859 augmented); adding weaker models degrades (All-9: 0.807).

Table I
Top Models by Variant

Model	Variant	MCC	Acc.
DeepStackEnsemble	Normal	0.8585	0.8937
HPO GRU	Normal	0.8524	0.8893
HPO TCN	Normal	0.8470	0.8843
CNN	Normal	0.8383	0.8774
HPO TCN	Augmented	0.8647	0.8982
HPO CNN	Augmented	0.8642	0.8977
Ensemble CNN+TCN	Augmented	0.8587	0.8932
RNN	Pure*	0.8808	0.9063
DS RNN_B	Noise*	0.7981	0.8458
HPO LSTM	Noise*	0.7908	0.8417

*Seed 3.

D. Per-Class Performance

Per-class F1 reveals consistent difficulty patterns. AB (alighting) leads (F1 0.893–0.925) due to distinctive descending trajectory; AA (inside) is hardest (0.831–0.894) from signal saturation near the AP; BB (at stop, 0.818–0.907) benefits from low RSSI; BA (boarding, 0.838–0.914) has moderate difficulty. Noise degrades AB most because concurrent near-AP transmissions mask its 20 dBm descent.

Figure 2 shows confusion matrices for the two best single-classification approaches. DeepStackEnsemble classifies 89.3% correctly, with AA↔BA (both near-AP states) and AB↔BB (both exterior locations) as primary confusion pairs. CNN shows a qualitatively similar pattern with more off-diagonal errors, particularly misclassifying AA as AB near the AP.

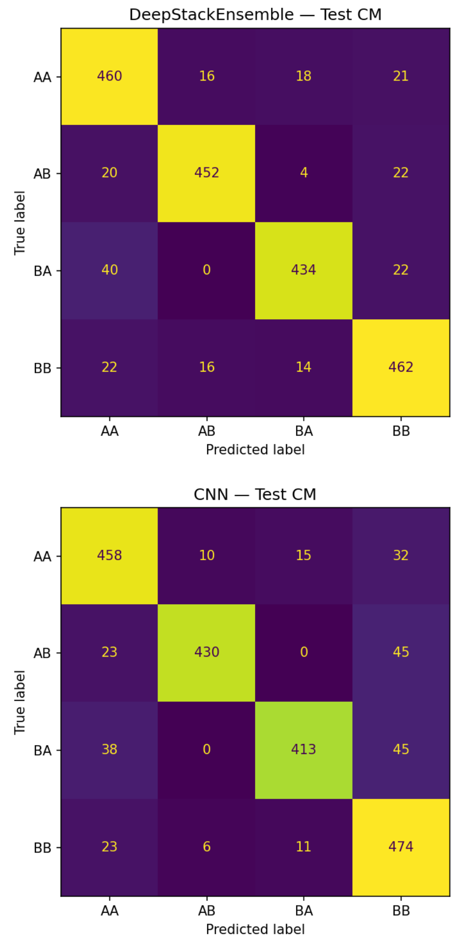

Figure 2. Confusion matrices: DeepStackEnsemble (left, MCC 0.859) and CNN (right, MCC 0.838).

E. Noise and Interference Analysis

Figure 3 quantifies mean RSSI shift per (Route × Interferer) pair. Three patterns emerge: (1) AA dominates as interferer (+45–53 dBm) because devices near the AP transmit at high power; BB causes minimal shift ($\leq$ +3.1 dBm). (2) AB is most affected (+46.6 dBm from AA): near-AP transmissions mask its descent. (3) Self-interference is minimal (+1.2–4.6 dBm).

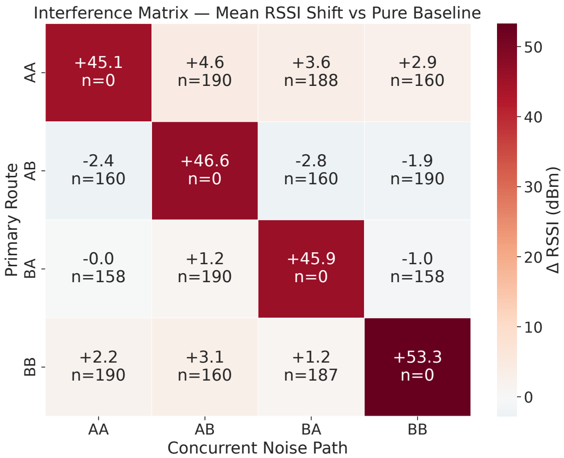

Figure 3. Interference heatmap: AA dominates; self-interference minimal.

Figure 4 explains *why* each class degrades differently under interference. AA suffers from BA raising its stable high-RSSI profile, making near-AP states hard to separate. AB's 20 dBm descent is flattened by AA into a constant profile indistinguishable from AA—directly explaining AB's noise-induced F1 drop. BA's ascending trajectory is compressed by AA interference, reducing the distinctive 15 dBm ascent. BB's flat -50 dBm pattern shifts upward under AA but retains its shape, explaining BB's noise resilience: classification depends on temporal flatness, not absolute level.

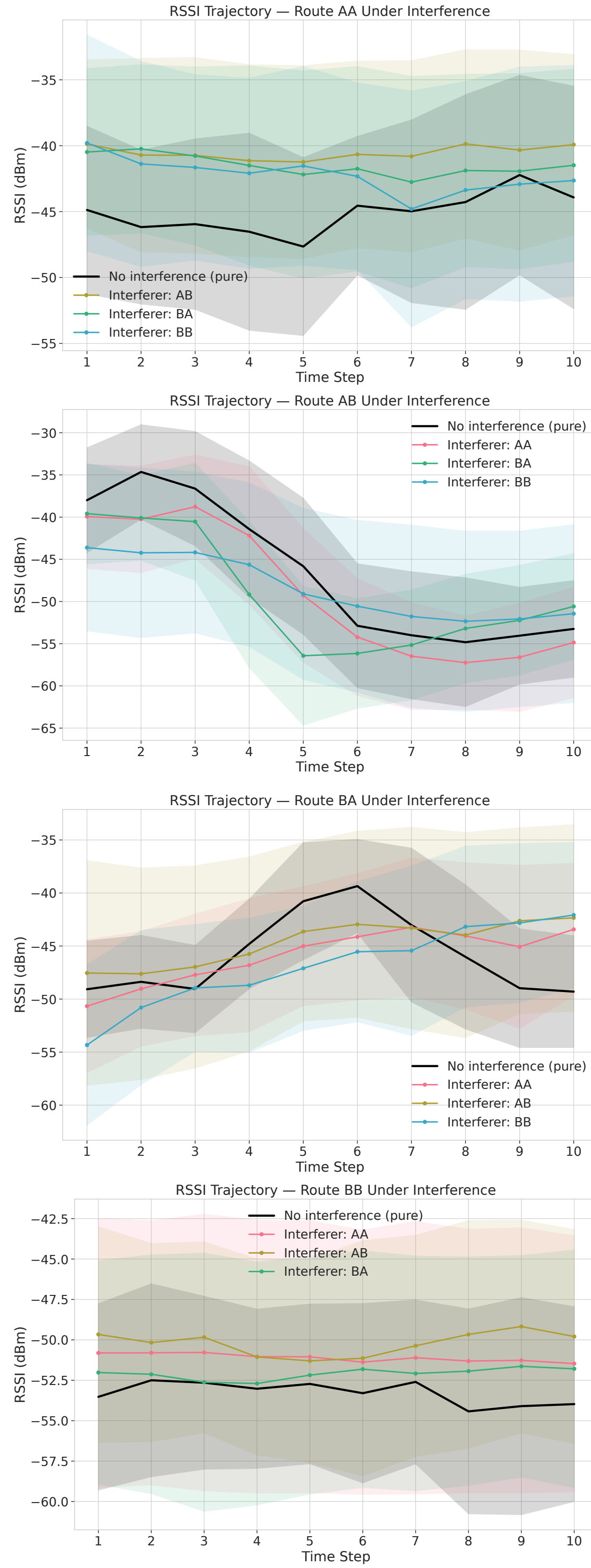

Figure 4. Trajectories under interference. Top-left: AA—BA raises profile. Top-right: AB—AA masks descent entirely. Bottom-left: BA—AA compresses ascent. Bottom-right: BB—flat pattern persists despite shift.

Per-timestep analysis reveals perturbation peaks at timesteps 4–6 (closest to AP). MetaFusion_GRU reaches MCC 0.783 on noise (vs. 0.758 for standard GRU) by encoding noise context, though the gap to pure (0.881) confirms that metadata alone cannot fully compensate for signal ambiguity under concurrent transmissions.

CNN (28K params, 129 s) offers the best efficiency-accuracy tradeoff among single models. Mamba (89K, 374 s) provides $O(L)$-complexity competitive accuracy. HPO GRU (198K, 107 s) is the strongest single model. DeepStackEnsemble (1.2M) suits server-side deployment.

F. Feature Importance

Permutation importance analysis across the 4 engineered channels and 10 timesteps, averaged over CNN, TCN, and GRU, reveals that window deviation—how far each timestep deviates from the per-sample mean—is the most informative channel (mean ΔMCC 9.1×10^{-3}), particularly at early timesteps ($t = 1$–3) where it captures each class's characteristic offset from its baseline. This confirms that trajectory shape, not absolute magnitude, carries the strongest discriminative signal.

Velocity (Δ) peaks at $t = 4$ (18.4×10^{-3}), the mid-trajectory moment of maximum movement. Raw RSSI matters most at $t = 6$, where absolute levels separate near-AP from exterior positions. Acceleration (Δ^2) contributes at $t = 3$–5, capturing directional movement onset. These results validate the 4-channel engineering strategy and align with the trajectory analysis in Figure 4, where AB's defining 20 dBm descent occurs during timesteps 4–7.

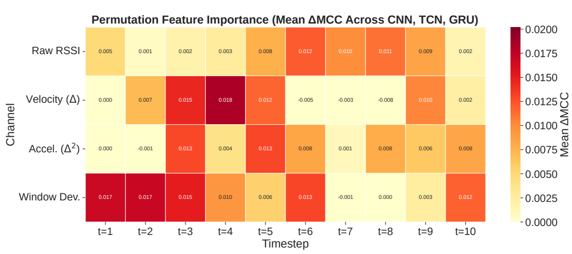

Figure 5. Permutation feature importance (ΔMCC) across 4 channels $\times$ 10 timesteps, averaged over CNN, TCN, and GRU.

V. Conclusion

This paper presented a comprehensive evaluation of deep learning architectures for RSSI-based passenger movement classification in public transport. Building on classical ML baselines (MCC $= 0.756$), we evaluated 11+ architectures spanning recurrent, convolutional, transformer, state-space, and mixture-of-experts models, supported by class-conditioned data augmentation and exhaustive hyperparameter optimization across four dataset variants.

The two-stage DeepStackEnsemble achieved MCC $= 0.859$ and accuracy 89.4% on combined data, a 13.6% relative improvement. State-space models (HPO Mamba, up to MCC $= 0.850$) demonstrated competitive performance with $O(L)$ complexity, establishing them as a viable alternative to transformers for resource-constrained deployment.

Cross-route interference analysis revealed a 9.4% absolute MCC degradation under multi-device noise. Heatmap analysis identified AA as the dominant interferer and AB as the most vulnerable route. Trajectory visualization traced the mechanism: concurrent near-AP transmissions erase AB's characteristic 20 dBm RSSI descent, the primary discriminative feature for alighting detection. BB proved most noise-resilient because its classification signal—temporal flatness—survives interference-level shifts intact.

Future work should explore attention mechanisms that learn to down-weight interfered timesteps, multi-AP fusion to disambiguate near-AP states through spatial diversity, and integration of movement classification with origin-destination matrix inference for end-to-end passenger flow modeling.

VI. Generative AI Declaration

During the preparation of this manuscript, the authors used DeepSeek as a writing assistant for language improvement, grammar correction, and formatting. All technical content, experimental design, model architectures, results, analysis, and conclusions represent the authors' original intellectual contribution. The authors take full responsibility for the accuracy and integrity of the content. No AI-generated figures, fabricated citations, or manipulated results are present in this work.

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